

Anderson–Rubin Test: First Stages and the Baseline Screen

Companion to the May 21, 2026 slides

1 The three regressions on the slides

Every cell on the slides comes from a cluster-robust regression at the municipality–year level, with municipality and year fixed effects α_m, δ_t and standard errors clustered by municipality. The deck reports three kinds of regression, each run at two margins: the four BNDES aggregate sectors, and BNDES sector $\times$ firm size (12 groups).

(a) First stage of the sector shares (slides 4–5). The endogenous object is the vector s_{mt} of sector employment shares. For an instrument stack $\mathcal{S}$ (a set of channels, [section 2](#)), each share is projected on the stack’s full instrument block:

$$s_{jmt} = \alpha_m^{(j)} + \delta_t^{(j)} + \Pi_j' Z_{mt}^{\mathcal{S}} + \Phi_j' \text{EC}_{mt}^{\mathcal{S}} + \mu_j \text{Vol}_{mt} + \nu_{jmt}. \quad (1)$$

This is the first stage embedded in the AR-test IV system; EC and Vol are included controls. Reported: the Sanderson–Windmeijer F for each share and the Kleibergen–Paap rank statistic of the block ([section 4](#)).

(b) First stage of the volume control (slide 6). The volume ratio is regressed, one channel c at a time, on that channel’s muni-aggregated instrument $\bar{Z}_{mt}^c = \sum_j Z_{jmt}^c$:

$$\text{Vol}_{mt} = \alpha_m + \delta_t + \pi_c \bar{Z}_{mt}^c + \Phi_c' \text{EC}_{mt}^c + \nu_{mt}. \quad (2)$$

Reported: the cluster-robust partial F on $\bar{Z}^c$. Channels clearing the 5% gate are the volume instruments used in treatment (iii) below.

(c) The AR test — baseline screen (slides 7–8). The AR reduced form, run once per excluded stack $\mathcal{S}$:

$$\log \text{GDP}_{mt} = \alpha_m + \delta_t + \gamma' Z_{mt}^{\mathcal{S}} + \psi' \text{EC}_{mt}^{\mathcal{S}} + (\text{volume treatment}) + \eta_{mt}. \quad (3)$$

This is the GDP equation with s_{mt} substituted out through its first stage (1). The reported **AR joint** F is the cluster-robust Wald statistic for $\gamma = \mathbf{0}$ — the joint significance of the stack’s instruments *only*; $\text{EC}^{\mathcal{S}}$ and the volume term are partialled out, not tested.

2 The seven channels and the excluded-stack design

Each alignment channel c pairs a set of political offices with an alignment shock Align_{mpt}^c ; the cross-office shocks are products of the tier-specific alignment indicators.

Channel	Offices	$\text{Align}_{mpt}^c = 1$ when party $p...$
M	mayor	holds the mayoralty of m
G	governor	is in the governor's coalition of $s(m)$
P	president	is in the presidential coalition
M·P	mayor, pres.	holds the mayoralty <i>and</i> is in the presidential coalition
M·G	mayor, gov.	holds the mayoralty <i>and</i> is in the governor's coalition
G·P	gov., pres.	is in the governor's <i>and</i> the presidential coalition
M·G·P	mayor, gov., pres.	is aligned at all three tiers simultaneously

A **stack** $\mathcal{S}$ is a set of channels. Equations (1) and (3) enter the full per-sector instrument block Z^c of *every* channel in $\mathcal{S}$ as excluded instruments: $Z_{mt}^{\mathcal{S}}$ stacks those blocks, $\text{EC}_{mt}^{\mathcal{S}}$ the matching exposure controls. Thus $\{M, P, M \cdot P\}$ means Z^M , Z^P , and $Z^{M \cdot P}$ are all excluded instruments at once. Eighteen stacks are evaluated:

- **Singletons** (7): each channel alone.
- **Mayor stacks** (4): $\{M, M \cdot P\}$, $\{M, M \cdot G\}$, $\{M, M \cdot G \cdot P\}$, $\{M, M \cdot P, M \cdot G, M \cdot G \cdot P\}$.
- **Parent pairs** (3): $\{M, G\}$, $\{M, P\}$, $\{G, P\}$.
- **Parent + interaction** (4): $\{M, P, M \cdot P\}$, $\{M, G, M \cdot G\}$, $\{G, P, G \cdot P\}$, $\{M, G, P, M \cdot G \cdot P\}$.

3 The three volume treatments

The volume term in (3) is toggled across three treatments.

Slide label	Volume term in (3)	Role
No vol.	—	volume left uncontrolled
Vol. control	λVol_{mt} , included	volume as a predetermined control
Vol. instr.	λVol_{mt} , instrumented	Full IV: Vol_{mt} instrumented

Under *Vol. instr.*, Vol_{mt} is instrumented by $\bar{Z}^c$ of the volume-relevant channels from (2) that are *not* in the stack $\mathcal{S}$ — using a channel already in $\mathcal{S}$ would make $\bar{Z}^c$ collinear with that stack's excluded Z^c block.

4 Diagnostics

- **Sanderson–Windmeijer** F — the cluster-robust first-stage F for one endogenous share, conditional on the other shares. “Identified shares” counts shares with SW $F \geq 10$.
- **Kleibergen–Paap rank statistic** — the cluster-robust rank test for the stack's instrument block, the relevance diagnostic for a vector-valued endogenous object.
- **AR joint** F — the cluster-robust Wald test of $\gamma = \mathbf{0}$ in (3), computed on the stack's instrument coefficients only.

5 Variable construction

Outcome — $\log \text{GDP}_{mt}$. Municipal log GDP (2018 prices), 2002–2017.

Endogenous shares — s_{jmt} . Sector j 's share of municipality m 's formal employment in year t , from RAIS. The simplex hold-out sector is the omitted base, leaving $J - 1$ endogenous shares.

Instrument — Z_{jmt}^c (**muni-relative owner share**). $Z_{jmt}^c = \sum_p \tilde{w}_{jmp,t}^c \text{Align}_{mpt}^c$, one column per retained sector. The weight

$$\tilde{w}_{jmp,t}^c = \frac{\sum_{s \in T_t^c} \sum_{f \in \mathcal{F}(j,m)} L_{f,p,s}}{\sum_{j'} \sum_{p'} \sum_{s \in T_t^c} \sum_{f \in \mathcal{F}(j',m)} L_{f,p',s}}$$

is cell (j, m) 's share of municipality m 's pre-window affiliated owner-years at party p ; $L_{f,p,s}$ counts owners of firm f affiliated with party p in year s (unaffiliated owners dropped), and $\mathcal{F}(j, m)$ is the set of firms ever active in sector j of m . The window $T_t^c = [e_c^{\min}(t) - 4, e_c^{\min}(t) - 1] \cap [2002, 2017]$ closes before any office in channel c can turn over, with $e_c^{\min}(t)$ the most recent election — among the channel's offices — on or before t .

Exposure control — $\text{EC}_{jm,t}^c$. The same weights summed over parties without the shift, $\text{EC}_{jm,t}^c = \sum_p \tilde{w}_{jmp,t}^c$: cell (j, m) 's share of the municipality's affiliated owner mass. It absorbs the share-component confounding of the shift-share instrument; one column per retained sector per channel.

Aggregate channel instrument — $\bar{Z}_{mt}^c$. $\bar{Z}_{mt}^c = \sum_j Z_{jmt}^c$, the channel's instrument summed over sectors. It is the scalar instrument for Vol_{mt} in the volume first stage (2) and in the Full-IV treatment.

Volume control — Vol_{mt} . $\text{Vol}_{mt} = \text{BNDES_total}_{mt} / \text{GDP}_{m,2002}$: current-year total municipal BNDES disbursements over initial-period (2002) municipal GDP. The denominator is fixed pre-treatment, so Vol_{mt} varies in t only through the numerator.